

**A GLM Analysis and Decomposition of China's Gender Wage
Differential: Impact of the COVID-19**

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Abstract

This study investigates the chronic change in and decomposition of the gender wage differential in China. The cross-sectional social survey data covering years 2017 and 2021 from the Chinese National Survey Data Archive is used for the analysis. A generalized linear model is used to obtain the extent of the gender wage differential, and the Oaxaca decomposition method with logarithm improvements is used for its decomposition. Results show that female workers earned 38 log points less than male workers in 2017, and 35 log points less in 2021. In terms of decomposition, the unexplained part, or the discriminatory part, takes up around 73% of the total differential in the 2017 data, and this number increases to 81% in 2021. The decomposition statistics are consistent with multiple China-based gender wage differential studies (Almas & Su, 2015; Elisabetta & Rong, 2012; Liu et al., 2000). Overall, this paper concludes that China's gender wage differential, although with a slight decreasing trend, is large and persistent; additionally, it is likely that an increasing part of the differential is due to discrimination.

A GLM Analysis and Decomposition of China's Gender Wage Differential: Impact of the COVID-19

Gender wage differential, or gender wage gap, refers to the earning differences between male and female workers. This topic has gained much research interest since the 1970s, as efforts to understand and narrow this gap are crucial for the development of a more equal and harmonious world. Although the gender wage differential generally has a decreasing trend, as many studies conclude, the gap is still persistent, as the US's female-to-male earnings ratio was 0.823 in 2019 (Semega et al., 2020). However, China's gender wage differential is understudied, as the latest comprehensive research on the gender wage differential was conducted in 2015 (Heshmati & Su, 2015). Moreover, the 2019 COVID strike could potentially affect China's gender wage differential due to its differential and significant impact on industries and occupations. Therefore, this paper aims to conduct a chronic analysis of China's wage differential by first estimating the wage differential using a generalized linear model and then decomposing the wage differential using the Oaxaca decomposition technique in both 2017 (pre-COVID period) and 2021 (post-COVID period). The results of this study will contribute to better understandings of the factors affecting China's gender wage differential, as well as finding the trends in China's gender wage gaps and gender discrimination.

Literature Review & Research Questions

Existing works of literature about gender wage differential yield mostly consistent conclusions about the worldwide existence of gender wage gaps. Verena and

Rajeev (2016) found prevalent gender wage differentials in both private and public economic sectors in Mauritius, and the differences in wages are larger at the bottom of the wage distribution; Semega et al. (2019) stated that the average male workers in the United States earned 48,769 dollars in 2019, while female workers earned only 35,826 dollars; Almas and Su (2015) also found that in China, an average female worker's earnings correspond to only 82.5% of the male worker's average earnings. Generally, though, the gender wage differential seems to be decreasing. Blau and Khan (2017) conducted a long-run trend analysis of US's the gender wage gap and found that the female-to-male wage ratio has improved from 0.6 in the 1970s to 0.79 in 2014.

In terms of the decomposition of the gender wage differential, most existing studies show that discrimination contributes significantly to the gender wage differential across countries in the past 20 years. In UK and Germany, only 23% to 40% of the gender wage differential could be explained by endowment differences (Stephen & Patrick, 2003); Geleta (2021) used Oaxaca decomposition and found that in the United States, after controlling for individual characteristics, the unexplained part of the gender wage differential is even larger, reaching 109% of the original gap, suggesting serious discrimination against women.

In China, when the opening up and reform policy was implemented, marketization of the economy led to larger gender wage differentials as human capital accumulation differed across genders and were more appropriately rewarded (Liu et al., 2000). Liu et al. (2000) also found that the discrimination effect was very large, taking up around 93% of the wage differential. In the 2010s, the gender wage gap was still

significant, and the unexplained part declined slightly, taking up 79% to 87% of the total wage differential (Almas & Su, 2015). Elisabetta and Rong (2012) also had similar findings, suggesting that discrimination contributes to 57% to 81% of the total gender wage differential across the wage distribution. These results suggest that discrimination against female workers is likely to persist in China, but the trend of the extent of discrimination is not clear.

These conclusions, however, received no further validation from more recent studies, especially after the COVID-19 pandemic, which could well jeopardize these conclusions due to the pandemic's differential impact on industries. For example, COVID-19 greatly boosted the development of video streaming platforms, which could potentially narrow the gender wage gap by increasing the income of female social media influencers. Therefore, this paper proposes the following research questions:

1. How large is China's gender wage differential before and after the COVID-19?
2. To what extent is China's gender wage differential due to discrimination? Is it different before and after the COVID-19?
3. What factors contribute to China's gender wage differential? Do they change over time?

Data and Methodology

Data

The data used in this study is the cross-sectional data from the Chinese National

Survey Data Archive (CNSDA). Maintained by the National Survey Research Center, this cross-sectional data is one of the most comprehensive social surveys in China, including both individual earnings and detailed characteristics. I chose the data in 2017 and 2021 to represent the pre-COVID era and post-COVID era respectively, since the latest data disclosed are from 2021. In 2017, a total of 14,793 observations were recorded, among which 6,680 were employed and thus used in the analysis; In 2021, a total of 8,148 observations were recorded, and 3,761 employed workers were analyzed. Since this survey is conducted by the same organization, based on similar samples, and relatively large in scale, it is assumed that the sample is representative in both years. The descriptive statistics of the observations used are summarized in the following table.

Table 1

Means and Distribution of Individual Characteristics

	2017		2021	
	Male	Female	Male	Female
Observations	3,655	3,025	1,993	1,768
Labor Income	50979.54	35,383.36	64770.79	44106.74
Tenure (years)	8.96	6.99	14.86	11.73
Experience (years)	12.59	9.41	19.31	15.90
Fulltime Job	0.93	0.93	0.93	0.93
Region (percentage)				
Shanghai	6.02	5.75	0.00	0.00
Yunnan	3.47	3.44	0.00	0.00
Inner Mongolia	0.82	0.79	1.86	1.30
Beijing	8.45	8.03	8.28	7.69
Jilin	3.86	4.26	0.00	0.00
Sichuan	4.76	5.39	0.00	0.00
Tianjin	2.46	2.25	0.00	0.00
Ningxia	0.74	1.09	0.75	1.81
Anhui	3.75	4.93	5.07	5.88

Shandong	5.39	5.09	7.53	7.30
Shanxi	2.46	2.08	2.86	3.79
Guangdong	4.24	5.42	0.00	0.00
Guangxi	3.26	3.74	5.22	5.20
Jiangsu	4.32	3.44	5.52	6.73
Jiangxi	4.46	4.20	8.93	4.36
Hebei	2.30	2.71	4.11	3.73
Henan	4.35	5.55	4.37	5.49
Zhejiang	4.27	3.87	6.12	6.73
Hubei	5.03	4.96	5.82	7.18
Hunan	3.61	3.17	4.06	4.86
Gansu	1.92	1.65	4.57	3.28
Fujian	2.35	2.25	4.37	2.77
Guizhou	2.71	2.15	0.00	0.00
Liaoning	3.15	3.01	4.57	3.96
Chongqing	2.44	2.48	10.59	12.16
Shanxi	3.75	4.00	5.42	5.77
Qinghai	0.98	0.66	0.00	0.00
Heilongjiang	4.68	3.64	0.00	0.00
Education (percentage)				
No education	4.52	13.93	3.72	11.22
private school	0.33	0.26	0.25	0.17
Primary school	19.46	20.74	18.61	20.11
Junior high school	31.7	24.08	31.14	27.65
Vocational high school	1.92	1.36	2.26	1.76
Regular high school	12.76	9.43	13.63	7.2
Technical secondary school	4.13	3.84	5.68	4.31
Technical school	0.93	0.30	0.65	0.11
College (Adult)	3.59	3.44	3.77	5.04
College (Regular)	6.46	7.11	5.73	5.5
University (Adult)	3.18	3.61	3.02	4.25
University (Regular)	9.17	9.82	9.91	10.82
Graduate and Above	1.86	2.08	1.61	1.87
Job Characteristics (percentage)				
Agricultural worker	28.78	33.09	29.57	32.00
Manual labourer	11.04	7.32	14.68	9.81
Technical labourer	12.44	4.04	15.89	9.81
Service worker	4.47	8.51	8.93	13.41
Junior office worker	6.11	7.72	5.70	10.33
Mid-level manager	2.93	1.59	0.96	0.80
Senior manager	0.33	0.36	0.10	0.11
junior technician	2.88	2.32	8.02	6.96
mid-level technician	1.81	1.56	1.41	1.20
Senior technician	0.58	0.60	0.50	0.40

Private company owner	0.77	0.26	0.15	0.23
Self-employed	7.54	8.02	5.20	5.08
Military/Police	1.23	0.30	3.08	0.11
Student	13.51	14.34	0.05	0.80
Retired	0.30	0.46	0.10	0.00
Freelancer	1.86	4.97	0.35	0.46
Others	3.43	4.54	5.30	8.50
Marriage (percentage)				
Married	84.52	87.94	83.29	86.6
Unmarried	13.21	9.65	13.55	10.18
Divorced	2.27	2.41	3.16	3.22

As shown in Table 1, in 2017 male workers earned on average 15596.18 yuan more than female workers, which gives a female-to-male earning ratio of 0.694; in 2021, both male and female workers earned more, and the female-to-male earning ratio is 0.681, which is slightly lower than that of 2017.

In 2021, both male and female workers have more experience. This could be explained by the layoff of less experienced workers under the economic pressures of COVID, which could be further validated by a similar increase in tenure in 2021. This could potentially affect the gender wage differential. In both years, more women (11-14 percent) than men (3-4 percent) received no education, but the proportion of women and men who had higher education are similar. In terms of job characteristics, more female workers than males work in the agricultural sector, which could be explained by women's lower basic education rates. In both years, there are more male labourers than female labourers. Notably, in 2017 the proportion of students is much higher, which is due to imperfect measurement: in 2017, the survey asked for job characteristics ten years ago. It is unclear why the survey was designed this way, but since there are no better job-related indicators, I have to assume that one's job characteristics ten years

ago are to some extent representative of his or her current job features. This imperfection was corrected in 2021. The marriage conditions of women and men are largely similar, with slightly more men being unmarried.

Methodology

1. Wage Differential: A Generalized Linear Model Approach

Traditionally, the wage differential is estimated by regressing wage on some control variables and a gender dummy variable. However, given that wages often exhibit skewed distributions, traditional linear regression is inappropriate. In this case, the median of wages is 24,000, while the mean of wages is 43,916, meaning the wage distribution is positively skewed. The Generalized Linear Model (GLM) allows for the use of distributions like the gamma distribution for the dependent variable, which can better capture the shape of wage data. Therefore, a GLM approach is employed in this paper using the logarithm transformation of the wage as the dependent variable:

$$\ln(W_{it}) = b_0 + b_{1t} * \text{female} + b_{it} * X_{it} + e_{it} \quad (1)$$

Where $\ln(W_{it})$ is the log annual labour income for worker i at year t , female is a dummy variable equaling to one if individual i is female. X_{it} is a set of control variables, and e_{it} is the residual. b_{it} indicates given other conditions, how much a female worker's earning differs from a male worker's. If b_{it} is proven significant, then females have significantly different wages from males. Specifically, I control for the following variables:

$$\ln(W_{it}) = b_0 + b_{1t} * \text{female} + b_{2t} * \text{Edu}_{it} + b_{3t} * \text{Job}_{it} + b_{4t} * \text{Area}_{it} + b_{5t} * \text{Full}_{it} + b_{6t} * \text{Exp}_{it} + b_{7t} * (\text{Exp}_{it})^2 + b_{8t} * \text{Marriage}_{it} + e_{it} \quad (2)$$

The specific interpretations of the variables in the above equation are shown in the following table. Since the correlation between experience and income is likely nonlinear, the Mincer earnings function is applied and both Exp_{it} and $(\text{Exp}_{it})^2$ are included.

Table 2

Description of variables in equation (2)

Variable	Description
Edu_{it}	A vector of dummy variables indicating individual i's education level at time t
Area_{it}	A vector of dummy variables indicating individual i's living region at time t
Job_{it}	A vector of dummy variables indicating individual i's job characteristics at time t
Exp_{it}	Represents the non-agriculture working experience (in years) of individual i at
$(\text{Exp}_{it})^2$	The square of experience
Full_{it}	A dummy variable equaling 1 if individual i has a full-time job
Marriage_{it}	A dummy variable equaling 1 if individual i is married at time t

2. Gender Wage Gap decomposition

The Oaxaca and Blinder (1973) decomposition technique will be used to determine wage differential magnitude and discrimination extent. Specifically, the following two equations are estimated separately:

$$\ln W_{mt} = a_{mt} + X_{mt} * \beta_{mt} + e_{mt} \quad (3)$$

$$\ln W_{ft} = a_{ft} + X_{ft} * \beta_{ft} + e_{ft} \quad (4)$$

Where W_{mt} is the wage of males at time t, and W_{ft} is the wage of females at time t. The log wages are calculated by taking the log of wage plus 0.001 in case some

workers' wages are equal to 0. X_{mt} and X_{ft} are the same set of control variables used in the GLM model above (apart from the gender dummy).

Then, to fit the decomposition into the GLM model, the wage differential D is defined as follows:

$$D = \frac{W_{ft}}{W_{mt}} \quad (5)$$

Lastly, the difference between (3) and (4), or the gender wage difference, could be taken and decomposed into the following equation:

$$\ln D = \ln \bar{W}_{ft} - \ln \bar{W}_{mt} = [(\bar{X}_{ft} - \bar{X}_{mt}) * \beta_{ft}] + [\bar{X}_{mt} * (\beta_{ft} - \beta_{mt}) + a_{mt} - a_{ft}] + e_{mt} - e_{ft} \quad (6)$$

Where $\bar{W}$ is the average wage, and $\bar{X}$ is the average characteristics. The characteristics effect, referring to the differences in male and female wages that can be explained by differences in endowments, is captured by $[(\bar{X}_{ft} - \bar{X}_{mt}) * \beta_{ft}]$; the discrimination effect, or the unexplained difference in wages, is captured by $[\bar{X}_{mt} * (\beta_{ft} - \beta_{mt}) + a_{mt} - a_{ft}]$. Although the discrimination effect could also capture the explanation effect of omitted variables, it is considered to represent the extent of discrimination against female workers, given almost all relevant factors are controlled. The data from 2017 and 2021 are examined respectively and the discrimination effects are calculated and compared.

Empirical Results

1. Results of the GLM Approach for Wage Differential

For clarity and concision, the most crucial results estimated using equation (2) are presented in Table 3. As shown in column 1, having controlled for job, region, and education, in 2017 an average female worker earned 37.8 log points less than an average male worker, and this number is statistically significant at the 1% level. In 2021, this number slightly decreases to 34.8 log points. This suggests a large, although decreasing gender wage gap.

Fulltime workers earn 16.5 log points higher than part-time workers in 2017, and 19.7 log points higher in 2021, which is significant and consistent with intuition. Experience has an inverted U shape with wage in both years, which is consistent with previous literature (Heshmati & Su, 2017; Zhang, 2004.) In 2017, married workers earned 14 log points more than unmarried workers, which could be explained by the higher age and more experiences of the married workers. However, this effect is insignificant in 2021.

Table 3
Estimation results of equation (2)

Logged Labor Income	2017		2021	
	Coefficient	Std. Err.	Coefficient	Std. Err.
Intercept	11.068***	(0.197)	11.703***	(0.309)
Female	-0.378***	(0.035)	-0.348***	(0.062)
Fulltime	0.165**	(0.066)	0.197**	(0.101)
Experience	0.050***	(0.005)	-0.001	(0.003)
Experience square	-0.001***	(0.000)	-0.00004**	(0.000)
Married	0.142**	(0.062)	0.103	(0.089)
Job controls	Yes		Yes	
Area controls	Yes		Yes	
Education controls	Yes		Yes	
Observations	6,680		3,761	

Note: *, **, and *** indicate significance at 10%, 5%, and 1% levels of significance respectively.

The detailed results for three vectors of controls can be found in Table 4. Unsurprisingly, the level of education has a significantly positive correlation with wages in both years, with the highest wages earned by workers with a university degree or above, and the lowest earned by the illiterate or workers with primary school education. Workers in relatively developed regions like Beijing and Shanghai are also much more likely to earn more, which is also reflected in the regression results. In terms of job characteristics, almost all non-agriculture workers earn more than agricultural workers in both years, and the surplus ranges from 14 log points to 103 log points. Among those, professional jobs like senior manager and private company owner are the most well-paid. This is consistent with previous literature (Heshmati & Su, 2017; Zhang, 2004.), and could be attributed to factors like better education and training opportunities, higher added values of the non-agricultural sector, and workers' self-selection. Notably, in 2021, some of the previously higher-paying jobs, like manual labourers and private company owners, have insignificant parameter estimates in 2021. This suggests external factors under COVID were likely to have severely increased the volatility of those workers' income. For example, many construction projects are paused or cancelled due to lockdowns, reducing the earnings of manual labourers.

Table 4*Detailed estimation of controls in equation (2)*

	2017		2021	
	Coefficient	Standard Error	Coefficient	Standard Error
Education				
No education	-1.716***	0.143	-2.552***	0.872
private school	-2.347***	0.336	-1.484***	0.232
Primary school	-1.459***	0.137	-1.177***	0.215
Junior	-1.200***	0.134	-0.918***	0.261
Vocational high school	-1.039***	0.179	-1.057***	0.222
Regular high school	-1.007***	0.135	-1.022***	0.229
Technical secondary	-0.914***	0.149	-1.480***	0.428
Technical school	-1.300***	0.240	-0.951***	0.228
College (Adult)	-0.675***	0.152	-0.758***	0.217
College (Regular)	-0.674***	0.136	-0.565***	0.229
University (Adult)	-0.506***	0.151	-0.412**	0.203
University (Regular)	-0.346***	0.130	-1.550**	0.272
Graduate and Above	(omitted)			
Region				
Shanghai	0.602***	0.111	0.000	0.000
Yunnan	-0.397***	0.121	0.000	0.000
Inner Mongolia	-0.213	0.219	-0.366	0.461
Beijing	0.277***	0.104	0.519**	0.202
Jilin	0.057	0.116	0.000	0.000
Sichuan	-0.572***	0.110	0.000	0.000
Tianjin	0.015	0.138	0.000	0.000
Ningxia	0.062	0.192	-0.026	0.334
Anhui	-0.099	0.115	-0.103	0.233
Shandong	-0.073	0.109	0.154	0.207
Shanxi	-0.312**	0.140	-0.251	0.233
Guangdong	0.345***	0.114	0.000	0.000
Guangxi	-0.426***	0.121	-0.304	0.239
Jiangsu	0.052	0.118	0.350*	0.209
Jiangxi	0.139	0.115	0.228	0.208
Hebei	-0.484***	0.136	-0.167	0.232
Henan	-0.647***	0.112	0.492**	0.229
Zhejiang	0.433***	0.118	0.582***	0.204
Hubei	-0.391***	0.112	-0.098	0.214
Hunan	-0.148	0.122	0.201	0.226
Gansu	-0.412***	0.148	-0.096	0.254
Fujian	0.164	0.138	0.314	0.234
Guizhou	-0.270**	0.136	0.000	0.000
Liaoning	-0.059	0.127	0.038	0.219
Chongqing	-0.271**	0.134	0.116	0.210
Shanxi	-0.278**	0.118	0.000	0.000
Qinghai	-0.234	0.198	0.000	0.000
Heilongjiang	(omitted)			

Job				
Agricultural worker	(omitted)			
Manual labourer	0.313***	0.071	0.144	0.123
Technical labourer	0.452***	0.076	0.417***	0.122
Service worker	0.529***	0.084	0.421***	0.129
Junior office worker	0.465***	0.094	0.279**	0.147
Mid-level manager	0.694***	0.130	0.762***	0.281
Senior manager	0.951***	0.293	0.730	0.749
junior technician	0.570***	0.120	0.246*	0.146
mid-level technician	0.540***	0.145	0.926***	0.248
Senior technician	0.914***	0.234	0.413	0.375
Private company owner	1.028***	0.234	0.515	0.568
Self-employed	0.756***	0.079	0.381***	0.148
Military/Police	0.371*	0.198	0.302*	0.231
Student	0.531***	0.079	-0.09	0.417
Retired	-0.083	0.279	0.000	0.000
Freelancer	0.318***	0.101	-0.725	0.676
Others	0.348***	0.102	0.240	0.155

Note: *, **, *** indicates significant at 10%, 5%, and 1% levels of significance respectively.

2. Results of Wage Differential Decomposition

Table 5 reports the results of wage differential decomposition. As discussed above, the total wage differential is decomposed into two parts, one indicating the differences in wages due to differences in characteristic endowments, or the explained part; the other is the unexplained part, which includes the discrimination effect and the effects of omitted variables. Although the effect of discrimination and omitted variables cannot be separated, the discrimination effect is believed to be dominant given all the controls.

Column 1 reports the decomposition in 2017. The unexplained part accounts for 73.2% of the total gap. This is consistent with the findings of Almas and Su (2015), and a bit smaller than the statistics in the study of Liu et al. (2000), with the former concluding that the unexplained part accounts for around 80% of the total gap, and the latter's conclusion is around 90%. This result is statistically significant at the 1% significance level and indicates that a large proportion of China's gender wage

differential is still due to discrimination.

To better understand the causes of the discrimination, the coefficients of the characteristics are presented under the 'Explained' row. Experience has an inverted U shape explaining power, which is consistent with the findings of Almas and Su (2015). This suggests that experience is likely to contribute to the discrimination, meaning experienced male workers are likely to be paid more than experienced female workers. Working full-time also enlarges the differential, but the effect is quite insignificant. Education has the greatest explaining power, taking up around 41% of the total explained part. This implies that a large part of the difference in male and female workers could be explained by their differences in education. Interestingly, this result is quite the opposite of the negative parameter presented in the study of Almas and Su (2015), which suggested that education enlarges the gender wage differential. The results in 2017, therefore, could be seen as an improvement in equality in the education system. Job characteristics also take up around 30% of the explanation, indicating female workers generally have lower-paying jobs. Marriage tends to enlarge the gap, since married women are likely to have less time and effort for work due to housework and babysitting, and are likely to be paid less than married men due to the above reasons.

Column 2 represents the decomposition in 2021. The total differential is smaller in 2021, which is consistent with the estimation results of the GLM equation, indicating that female workers' wages are relatively higher in 2021 than in 2017. However, the unexplained part takes up 81.7% of the wage differential, which is an 8.5% increase. This coefficient is statistically significant at the 5% level. Looking specifically into the

factors, it could be observed that marriage, experience, and job characteristics exhibit similar patterns. Notably, the explaining power of education is much smaller. One explanation for this change could be that COVID forced almost all forms of education to be conducted online, which increased education inequality (since remote areas do not have sufficient resources for online education) and reduced education quality. This might decrease the return to education, especially for female workers. The explaining power of regions, on the other hand, increases. This could be attributed to the larger regional development gap resulting from the lockdowns.

Table 5

Wage Differential Decomposition

	2017	2021
Male	9.122	10.237
Female	7.746	9.222
Total Differential	1.376***	1.015***
Explained	0.368***	0.185**
Experience	0.648	0.019
Experience Square	-0.557	-0.013
Fulltime	-0.001	0.004
Job Characteristics	0.115	0.057
Marriage	-0.011	-0.013
Education	0.153	0.006
Region	0.021	0.125
Unexplained	1.008***	0.831***
Due to Endowments, %	26.7	18.3
Due to Discrimination, %	73.2	81.7

Note: only the significance level of the total differential, the explained part, and the unexplained part are displayed. *, **, *** indicates significant at 10%, 5%, and 1% levels of significance respectively.

Therefore, the conclusion of this section is that although the total gender wage differential in China is decreasing, persistent discrimination against female workers still exists in China, if not increasing, and contributes greatly to the gender wage differential

before and after the COVID-19 pandemic.

Discussion and Conclusion

This paper examines China's gender wage differential in the years 2017 and 2021, intending to understand how large and persistent the gap is, the extent to which the gap is due to discrimination, and the factors contributing to it. First, a generalized linear model is used to estimate the extent of China's wage differential, which found that an average female worker earns 38 log points less than an average male worker in 2017, and 35 log points less in 2021. This large but decreasing gender wage differential is consistent with the findings of Semega et al. (2020). Then, the wage differential is decomposed using the Oaxaca decomposition method, and it is found that in 2017, around 73% of the differential is due to discrimination, and this number increased to 81% in 2021. These numbers echo the findings of many China-based wage differential studies (Almas & Su, 2015; Elisabetta & Rong, 2012; Liu et al., 2000). Factors like education, region, and job characteristics explain the gap, while experience and marriage tend to enlarge the gap. Therefore, this paper concludes that China's gender wage differential, although with a decreasing trend, is significant and persistent, and a large and likely increasing part of the differential is due to discrimination.

This study, using the data from 2017 and 2021, successfully filled in the blank area where gender wage differential studies are missing in recent China, especially in the post-COVID period. This study extends the validity of the conclusions made in previous papers, verifying the existence of a persistent gender wage gap in China up to 2021. Crucially, an increasing trend in discrimination against female workers is

observed in the gender wage gap, and various factors contributing to the gender wage differential are identified.

Based on these findings, I propose the following policy suggestions to narrow the gender wage differential. Firstly, governments should propose efforts to eliminate gender discrimination in the labour market. China's labour laws are currently not detailed enough regarding gender issues, and more specific requirements like wage transparency and equal pay for equivalent work should be institutionalized. Other policies could also affect the gender wage differential by changing the factors contributing to it. Although factors like experience and marriage are not directly affected by policies, policies focusing on education and development equality could have a direct effect. Since it is found that education has a weakening explaining power, governments should foster the development of high-quality education, especially higher education, aligning education more with industry needs and shifting the role of education from pure signalling to increasing human capital. Additionally, governments should continue promoting common prosperity and support the development of less-developed areas. For instance, the government could relax the Hukou system, which could boost migration, reduce the rural-urban income gap, and mitigate the gender wage differential. Lastly, it is also vital to mitigate taste-based discrimination in society by promoting gender equality in public opinion.

Like any other regression-based studies, the GLM model applied in this study might suffer from omitted variable bias. However, given that this study controls for almost all traditionally significant factors contributing to wage differences, including

education, experience, working full-time, marriage, job, and region, the omitted variable bias is believed to be quite insignificant. One potential concern for the decomposition in this study could be the representativeness of the sample in the cross-sectional data. Since this study uses cross-sectional data, the effect captured is mainly cohort effects. In the 2017 data, the number of observations is 6,680, and in 2021, this number is only 3,761, and the two groups have different experiences. This suggests that this paper may be comparing the gender wage differential in all workers to the gender wage differential in the more experienced workers, which may not capture the whole picture. Yet if this change in the data is also the trend in the whole labour force, the estimate is still valid, and it is very likely the case: many inexperienced workers are laid off during the pandemic, increasing the average experiences of workers. Another data-related concern is that discrimination could lead to barriers that prevent female workers from working in certain industries, which makes controlling for industry underestimate the discrimination effect. Lastly, as mentioned in the data part, the job characteristics measurement in 2017 was not perfect. Therefore, future studies could focus on obtaining and analyzing panel data with better measurements, which better captures the change in the gender wage differential as well as discrimination within a certain group. Further studies could also look more closely into industries and forms more comprehensive conclusions by excluding female-unfriendly industry controls.

Despite the minor limitations, this study still yields consistent and significant conclusions about China's wage differential: although the total gender wage differential in China has slightly decreased, persistent discrimination against female workers still

exists, if not increasing, in China. The results of this study have important implications for both understanding the gender wage differential and promoting gender equality in the future.

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